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DAY-3

Configuring RIP Theory: RIP concepts: Hop count, updates, and convergence. Practical : Configure RIP on a multi-router network. Task: Capture RIP updates using simulation tools and analyze them.

Q-1 Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

[A]

- What is RIP?
  - RIP (Routing Information Protocol) is a dynamic routing protocol that automatically shares routing information between routers. It uses hop count as its routing metric to determine the best path.
    - **Routing Protocol Type:** Distance Vector
    - **Metric:** Hop Count
    - **Maximum Hop Count:** 15
    - **Hop Count 16:** Network is unreachable
    - **Update Interval:** Every 30 seconds
    - **Administrative Distance:** 120

- RIP Updates
  - RIP sends routing table updates every **30 seconds**.
  - Updates are sent using **UDP Port 520**.
  - RIP Version 2 supports:
  - Classless routing (CIDR)
  - VLSM
  - Authentication
  - Multicast address **224.0.0.9**

- Convergence
  - **Convergence** is the time taken for all routers to update their routing tables after a network change.

    - Example
      - A link between Router1 and Router2 fails. ○ RIP detects the failure. ○ Routers exchange updated routing information. ○ All routers learn the new route.
      - The network is now converged.

Device	Interface	IP Address	Subnet Mask
Router1	G0/0	192.168.1.1	255.255.255.0
Router1	S0/0/0	10.0.12.1	255.255.255.252
Router2	S0/0/0	10.0.12.2	255.255.255.252
Router2	S0/0/1	10.0.23.1	255.255.255.252
Router2	G0/0	192.168.2.1	255.255.255.0
Router3	S0/0/1	10.0.23.2	255.255.255.252
Router3	G0/0	192.168.3.1	255.255.255.0

- Configure RIP Version 2

   Router 1

```
enable configure
terminal

interface g0/0 ip address 192.168.1.1
255.255.255.0 no shutdown
exit

interface s0/0/0 ip address 10.0.12.1
255.255.255.252 clock rate 64000 no
shutdown
exit
```

```
router rip version 2
no auto-summary
network 192.168.1.0
network 10.0.12.0

end
```

Router 2

```
enable configure
terminal

interface g0/0 ip address 192.168.2.1
255.255.255.0 no shutdown
exit

interface s0/0/0 ip address 10.0.12.2
255.255.255.252 no shutdown
exit

interface s0/0/1 ip address 10.0.23.1
255.255.255.252 no shutdown
exit

router rip version 2
no auto-summary
network 192.168.2.0
network 10.0.12.0
network 10.0.23.0

end
```

- Router 3

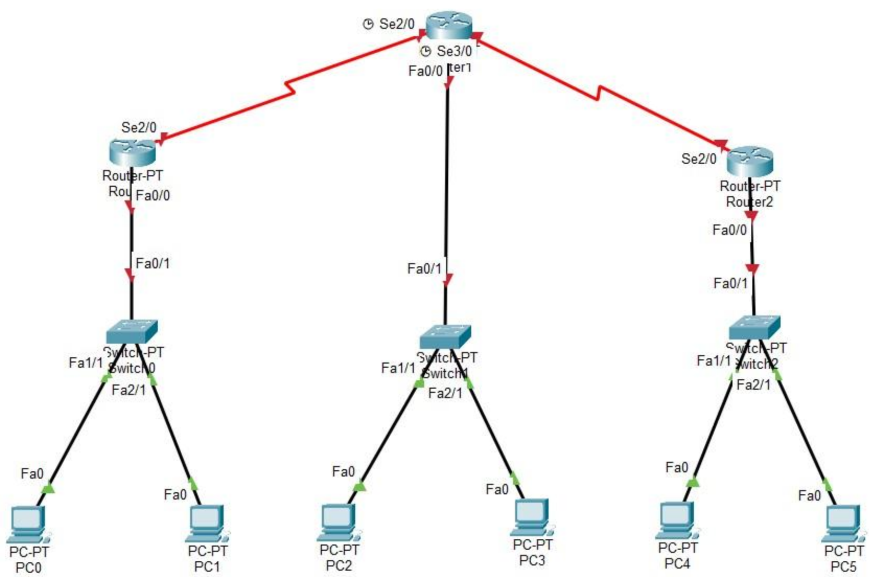
```
enable
configure terminal

interface g0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit

interface s0/0/1
ip address 10.0.23.2 255.255.255.252
no shutdown
exit

router rip
version 2
no auto-summary
network 192.168.3.0
network 10.0.23.0

end
    write
```



- Increase Hop Count

Original : R1 ------ R2 ------ R3
Increase : R1 ------ R2 ------ R4 ------ R3

- Example

router rip version
2 no auto-
summary
network 10.0.24.0
network 10.0.34.0

Q-2 Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers. Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

[A]:

- Capture RIP Updates in Simulation Mode
  1. Open **Cisco Packet Tracer**.
  2. Switch from **Real-Time** mode to **Simulation** mode.
  3. Click **Edit Filters**.
  4. Clear all protocol selections.
  5. Select only **RIP**.
  6. Click **Capture/Forward** or **Auto Capture/Play**.
  7. Observe the RIP packets exchanged between routers.
  8. Click on any RIP packet to inspect its details.

- Observation Table

Parameter	**Observation**
Protocol	RIP Version 2
Transport Protocol	UDP
Port Number	520
Destination Address	224.0.0.9
Update Interval	30 seconds
Metric Used	Hop Count
Maximum Hop Count	15
Hop Count 16	Network Unreachable